

Algebra II with Trigonometry

Chapter 6.2

Mental Math

Saved tutoring response with MathJax rendering. Original PDF file:
[4506907_alg-2trig-h-chp-6-2-note-docx.pdf](#)

Certainly! Here are 10 **mental math questions** based on the main concepts from your Algebra 2/Trig material on trigonometric functions of real numbers. These are meant to be answerable **without paper and pencil** and test conceptual understanding, not detailed calculations.

1. Unit Circle Coordinates

If a point on the unit circle has an x-coordinate of $1/2$, what is the cosine of the corresponding angle?

2. Quadrant Signs

Which trigonometric functions are positive in Quadrant III?

3. Function Domains

Is $\tan\left(\frac{\pi}{2}\right)$ defined? Why or why not?

4. Reciprocal Functions

If $\sin t = \frac{1}{2}$, what is $\csc t$?

5. Quadrant Identification

If $\sin t < 0$ and $\cos t > 0$, in which quadrant does terminal side of t lie?

6. Even-Odd Property

Is the cosine function even or odd?

7. Function Signs

If a point is in Quadrant II, is $\sec t$ positive or negative?

8. Pythagorean Identity

If $\sin^2 t = \frac{1}{4}$, what is $\cos^2 t$?

9. Reference Angle

What is the reference angle for 210° (or $\frac{7\pi}{6}$)?

10. Function Value through Symmetry

If $\sin(\alpha) = 0.6$, what is $\sin(-\alpha)$?

Let me know if you want brief explanations or answers for these!